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UNIT-II

GEOMETRICAL TRANSFORMATION

2D TRANSFORMATIONS

In many applications, there is a need for altering and manipulating displays. Many applications actually use the geometric transformation to change.

* the position
* the orientation
* size of objects

Basic geometric transformations are:

1) Translation
2) Rotation
3) Scaling

other transformations:

1) Reflection
2) Shearing.

TRANSLATION

* Translation means repositioning the object by changing its coordinates by adding some additional translation distances to original coordinate.

* We translate a 2D point by adding translation distances t_x and t_y, to the original coordinate position (x, y) to move point to a new position (x', y'). Fig (1)

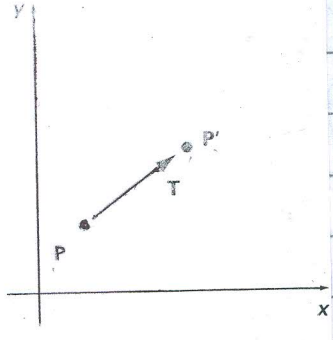


Fig: 1
Translating a point from position P to position P' with translation vector T.

$$x' = x + t_x \quad \text{---} \quad (1)$$
$$y' = y + t_y$$

The translation distance pair (t_x, t_y) is called a translation vector or shift vector.

We express the translation equations (1) as a single matrix equation as:

$$P' = \begin{bmatrix} x' \\ y' \end{bmatrix}, \quad P = \begin{bmatrix} x \\ y \end{bmatrix}, \quad T \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$

i.e

$$P' = P + T.$$

Translation is a rigid body transformation that move object from one position to another. (fig2)

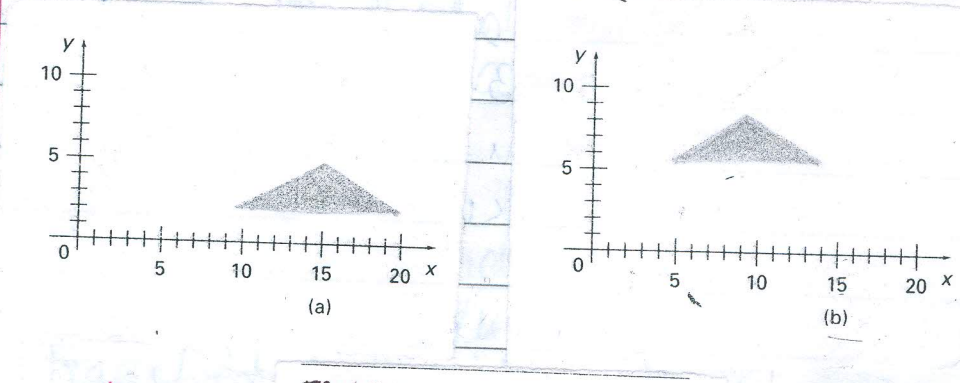


Fig 2.
Moving a polygon from position (a) to position (b) with the translation vector (-5.50, 3.75).

ROTATION

In rotation, we specify a rotation angle θ and the position (x_r, y_r) of rotation point (or pivot point) about which the object is to be rotated. (Fig 3)

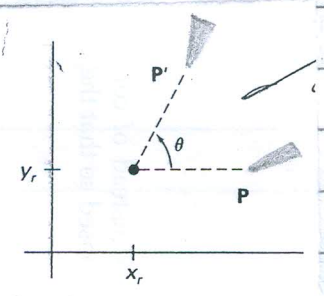


Fig:3
Rotation of an object through angle θ about the pivot point (x_r, y_r).

Positive values for rotation angle define counterclockwise rotations about the pivot point as in fig 3. and negative values for rotation angle define the clockwise rotations of objects.

This transformation can also be described as a rotation about a rotation axis that is perpendicular to the xy plane and passes through the pivot point.

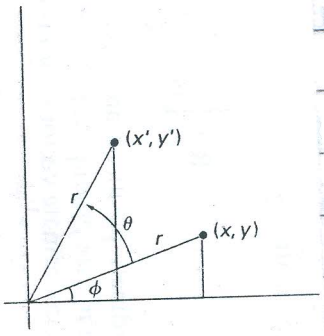


Fig: 4

Rotation of a point from position (x, y) to position (x', y') through an angle θ relative to the coordinate origin. The original angular displacement of the point from the x axis is ϕ.

We first determine the transformation equations for rotation of a point position position P when the pivot point is at the coordinate origin.

The angular and coordinate relationship of the original and transformed point positions are shown in fig 4.

In the fig 4, r is the constant distance of a point from origin, angle ϕ is original angular position of the point from the horizontal and θ is the rotation angle.

Using standard trignometric identities, we can express the transformed coordinates in terms of angles θ and ϕ as:

$$x' = r\cos(\theta+\phi)$$
$$= r\cos\phi\cos\theta - r\sin\phi\sin\theta$$

$$y' = r\sin(\phi+\theta)$$
$$= r\cos\phi\sin\theta + r\sin\phi\cos\theta$$

The original coordinates of the point in polar coordinates are

$$x = r\cos\phi, \quad y = r\sin\phi.$$

Substituting the above equations we get

the transformed equations for rotating a point at position (x, y) through an angle θ about the origin:

$$x' = x\cos\theta - y\sin\theta$$
$$y' = x\sin\theta + y\cos\theta \quad \text{—(2)}$$

We can write the rotation equations in the matrix form:

$$P' = R \cdot P$$

where

$$R = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

$$P = \begin{bmatrix} x \\ y \end{bmatrix}, \quad P' = \begin{bmatrix} x' \\ y' \end{bmatrix}$$

The transformation equations for rotation of a point about any specific rotation position (x_r, y_r) (Figs)

$$x' = x_r + (x - x_r)\cos\theta - (y - y_r)\sin\theta$$
$$y' = y_r + (x - x_r)\sin\theta + (y - y_r)\cos\theta$$

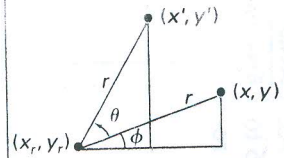


Fig: 5
Rotating a point from position (x, y) to position (x', y') through an angle θ about rotation point (x_r, y_r).

As with Translation, rotations are rigid body transformation that move objects without deformation. Every point on an object is rotated through the same angle.

SCALING

A scaling transformation alters the size of an object. This operation can be carried out for polygons by multiplying the coordinate values (x, y) of each vertex by scaling factors s_x and s_y to produce the transformed coordinates (x', y')

$$x' = x \cdot S_x$$
$$y' = y \cdot S_y \quad \text{— (3)}$$

The above equations can also be written in matrix form

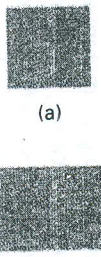
$$P' = S \cdot P$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} S_x & 0 \\ 0 & S_y \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix}$$

where S is the scaling matrix.

S_x, S_y values	effect
1) less than 1	Reduce the size of object (Close to coordinate origin)
2) greater than 1	increase the size of object.
3) equal to 1	size of object is unchanged.
4) $S_x = S_y$	uniform scaling is produced that maintains relative object proportion.
5) $S_x \neq S_y$	differential scaling used to design applications.

6) $S_x = 2, S_y = 1$ A square can be turned to rectangle. (fig 6)



(a)

(b)

Fig: 6
Turning a square (a) into a rectangle (b) with scaling factors $s_x = 2$ and $s_y = 1$.

OTHER TRANSFORMATIONS

REFLECTION

A reflection is a transformation that produces a mirror image of an object. The mirror image for a 2D reflection is generated relative to an axis of reflection by rotating the object 180° about the reflection axis.

Following are examples of some common reflections.

Case 1 - Reflection about the line $y = 0$ the x axis, is accomplished with transformation matrix as:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

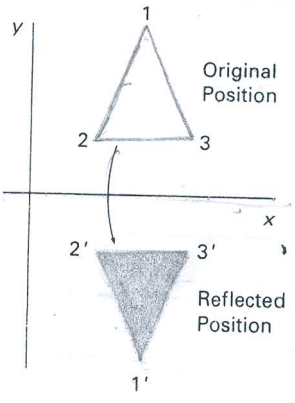


Fig: 7.
Reflection of an object about the x axis.

The resulting orientation of an object after it has been reflected about the x axis is shown in fig 7.

Case 2: Reflection about the line $x = 0$ the y axis is accomplished with

Transformation matrix as

$$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

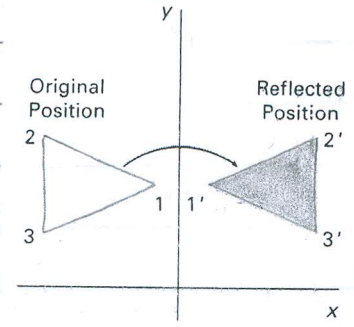


Fig: 8
Reflection of an object about the y axis.

The resulting orientation of an object after it has been reflected about the y axis is shown in fig 8.

Case 3 Reflection relative to the coordinate origin, has the matrix representation as:

$$\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

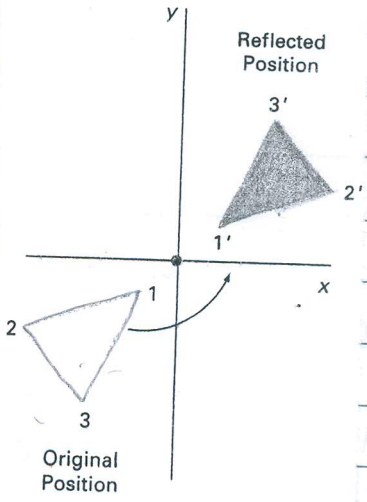


Fig: 9
Reflection of an object relative to an axis perpendicular to the *xy* plane and passing through the coordinate origin.

Case 4: Reflection of an object with repect to the line $y = x$. The matrix is

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

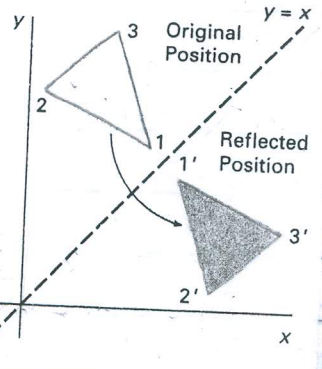


Fig: 10
Reflection of an object with respect to the line $y = x$.

Case:5 Reflection of an object with respect to line $y = -x$. The matrix is:

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

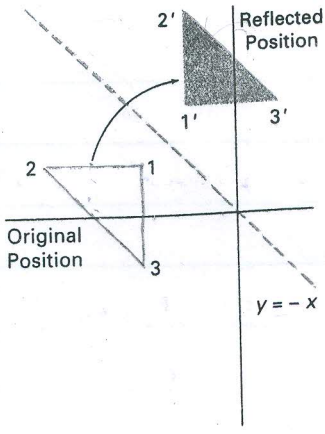


Fig 10
Reflection with respect to the line $y = -x$.

SHEAR

A transformation that distorts the shape of an object such that the transformed shape appears as if the object were composed of internal layers that had

been caused to slide over each other is called shear.

Case 1: An x-direction shear relative to the x-axis is produced with the transformation matrix.

$$\begin{bmatrix} 1 & sh_x & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

which transforms coordinate positions as

$$x' = x + sh_x \cdot y$$
$$y' = y$$

For eg $sh_x = 2$, changes the square into a parallelogram. (Fig 11)

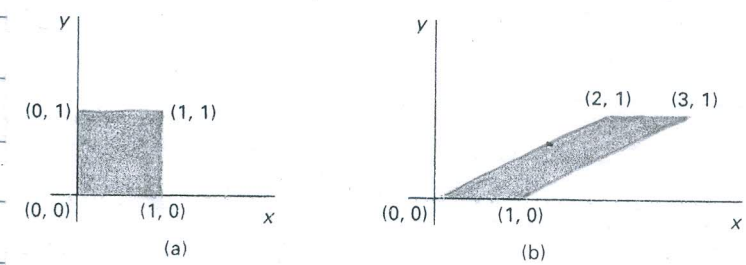


Fig: 11.
A unit square (a) is converted to a parallelogram (b) using the x-direction shear matrix) with $sh_x = 2$.

Case 2: x-direction shears relative to other reference lines $y = y_{ref}$

$$\begin{bmatrix} 1 & sh_x & -sh_x \cdot y_{ref} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

with coordinate positions transformed as:

$$x' = x + sh_x(y - y_{ref})$$
$$y' = 1$$

For eg fig 12 depicts the above shearing transformation with shearing parameter value $sh_x = 1/2$ relative to line $y_{ref} = -1$.

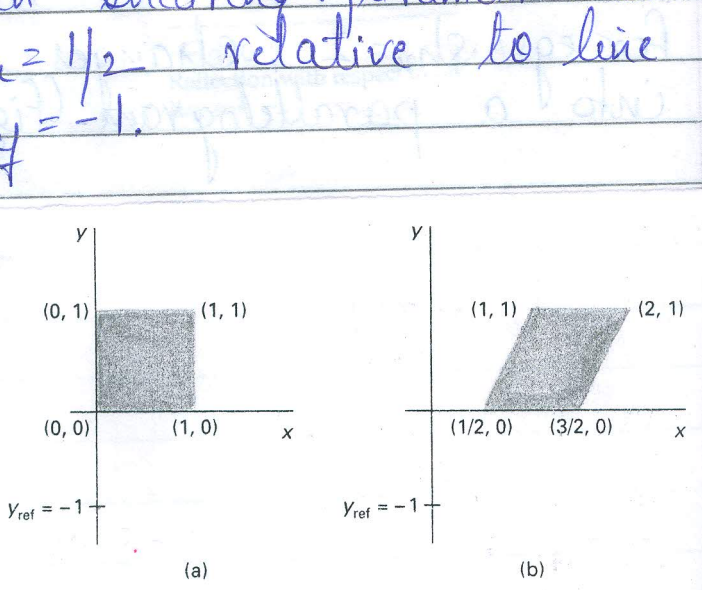


Fig 12:
A unit square (a) is transformed to a shifted parallelogram (b) with $sh_x = 1/2$ and $y_{ref} = -1$ in the shear matrix

Case 3 y-direction shear relative to the line $x = x_{ref}$ is generated with transformation matrix

$$\begin{bmatrix} 1 & 0 & 0 \\ sh_y & 1 & -sh_y \cdot x_{ref} \\ 0 & 0 & 1 \end{bmatrix}$$

which generates transformed coordinate positions

$$x' = x$$
$$y' = sh_y(x - x_{ref}) + y$$

For eg: Fig 13 illustrates the conversion of square into a parallelogram with $sh_y = 1/2$ and $x_{ref} = -1$.

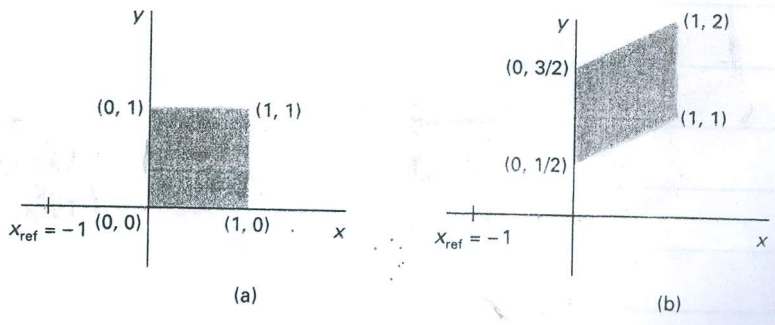


Fig: 13

A unit square (a) is turned into a shifted parallelogram (b) with parameter values $sh_y = 1/2$ and $x_{ref} = -1$ in the y-direction using shearing transformation

HOMOGENEOUS COORDINATES AND MATRIX REPRENTATION OF 2D TRANSFORMATION

* We have seen that each of the basic transformation can be expressed in the general matrix form i.e

$$P' = M_1 \cdot P + M_2$$

with coordinate positions P and P' represented as column vectors. Matrix M_1 is 2 by 2 array containing multiplicative factors, and M_2 is a two element column matrix containing translational terms.

* We can combine the multiplicative and translational terms for 2D geometric transformation into single matrix representation by (expanding

the 2 by 2 matrix representations to 3 by 3 matrices. This allows us to represent and express all transformation equations as matrix multiplications, providing that we also expand the matrix representations for coordinate positions.

To express any 2D transformation as a matrix multiplication, we represent each Cartesian coordinate position (x, y) with homogeneous coordinate triple (x_h, y_h, h), where

$$x = \frac{x_h}{h}, \quad y = \frac{y_h}{h}$$

For 2D geometric transformation, we can choose the homogeneous parameter h to be any nonzero value.

A convinient choice is simply to set $h = 1$. Each 2D position is then represented with homogeneous coordinate $(x, y, 1)$

Expressing positions in homogeneous coordinates allows us to represent all geometric transformation equations as matrix multiplications. Coordinates are represented with three element column vector and transformation operations are written as 3 by 3 matrices.

TRANSLATION - about coordinate origin

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

which can be written as

$$P' = T(t_x, t_y) \cdot P$$

The inverse of translation matrix is obtained by replacing the translation parameters t_x and t_y with their negatives i.e $-t_x$ and $-t_y$.

ROTATION - about coordinate origin

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

which can be written as:

$$P' = R(\theta) \cdot P$$

For inverse rotation θ is replaced with $-\theta$.

SCALING - about coordinate origin

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

or

$$P' = S(s_x, s_y) \cdot P$$

For inverse scaling replace s_x and s_y with $1/s_x$ and $1/s_y$.

COMPOSITION OF 2D TRANSFORMATIONS

We can set up a matrix for any sequence of transformation known as composite transformation matrix.

The basic purpose of composing transformation is to gain efficiency by applying a single composed transformation to a point, rather than applying a series of transformations, one after the another.

TRANSLATIONS

If two successive translation vectors (t_{x1}, t_{y1}) and (t_{x2}, t_{y2}) are applied to a coordinate position P, the final transformed location P' is calculated as:

$$P' = T(t_{x2}, t_{y2}) \cdot \{T(t_{x1}, t_{y1}) \cdot P\}$$
$$= \{T(t_{x2}, t_{y2}) \cdot T(t_{x1}, t_{y1})\} \cdot P$$

The composite transformation matrix for this sequence of translations is

$$\begin{bmatrix} 1 & 0 & t_{x2} \\ 0 & 1 & t_{y2} \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & t_{x1} \\ 0 & 1 & t_{y1} \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & t_{x1} + t_{x2} \\ 0 & 1 & t_{y1} + t_{y2} \\ 0 & 0 & 1 \end{bmatrix}$$

or

$$T(t_{x2}, t_{y2}) \cdot T(t_{x1}, t_{y1}) =$$
$$T(t_{x1} + t_{x2}, t_{y1} + t_{y2})$$

which demonstrates that two successive translations are additive.

ROTATIONS

Two successive rotations applied to point P produce the transformed position as:

$$P' = R(\theta_2) \cdot \{R(\theta_1) \cdot P\}$$

$$= \{R(\theta_2) \cdot R(\theta_1)\} \cdot P$$

The composite transformations matrix for this sequence of rotations is

$$\begin{bmatrix} \cos(\theta_2) & -\sin\theta_2 & 0 \\ \sin\theta_2 & \cos\theta_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} \cos\theta_1 & -\sin\theta_1 & 0 \\ \sin\theta_1 & \cos\theta_1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \cos\theta_2\cos\theta_1 - \sin\theta_2\sin\theta_1 & -\cos\theta_2\sin\theta_1 - \sin\theta_2\cos\theta_1 & 0 \\ \sin\theta_2\cos\theta_1 + \cos\theta_2\sin\theta_1 & -\sin\theta_2\sin\theta_1 + \cos\theta_2\cos\theta_1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \cos(\theta_2+\theta_1) & -\sin(\theta_2+\theta_1) & 0 \\ \sin(\theta_2+\theta_1) & \cos(\theta_2+\theta_1) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

using trignometric identities:

$\cos(A+B) = \cos A \cos B - \sin A \sin B$

$\sin(A+B) = \sin A \cos B + \cos A \sin B$

or

$$R(\theta_1) \cdot R(\theta_2) = R(\theta_1 + \theta_2)$$

which demonstrates that two successive rotations are additi

SCALINGS

Two successive scaling operations applied to point P produce the transformed positions as.

$$P' = S(s_{x_2}, s_{y_2}) \cdot \{S(s_{x_1}, s_{y_1}) \cdot P\}$$

$$= \{S(s_{x_2}, s_{y_2}) \cdot S(s_{x_1}, s_{y_1})\} \cdot P$$

The composition matrix for this sequence of scalings is

$$\begin{bmatrix} s_{x_2} & 0 & 0 \\ 0 & s_{y_2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} s_{x_1} & 0 & 0 \\ 0 & s_{y_1} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} s_{x_1} \cdot s_{x_2} & 0 & 0 \\ 0 & s_{y_1} \cdot s_{y_2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

or

$$S(s_{x2}, s_{y2}) \cdot S(s_{x1}, s_{y1}) = S(s_{x1} \cdot s_{x2}, s_{y1} \cdot s_{y2})$$

which demonstrates that two successive scalings are multiplicative.

GENERAL PIVOT POINT ROTATION

To rotate an object about a pivot point (arbitary point) (x_r, y_r), we perform following sequence of Translate - Rotate - Translate operations:

1) Translate the object so that the pivot point position is moved to the coordinate origin.

2) Rotate the object about the coordinate origin

3) Inverse Translate the object so that the pivot point

is returned to its original position.

The transformation sequence is illustrated in Fig 1.4.

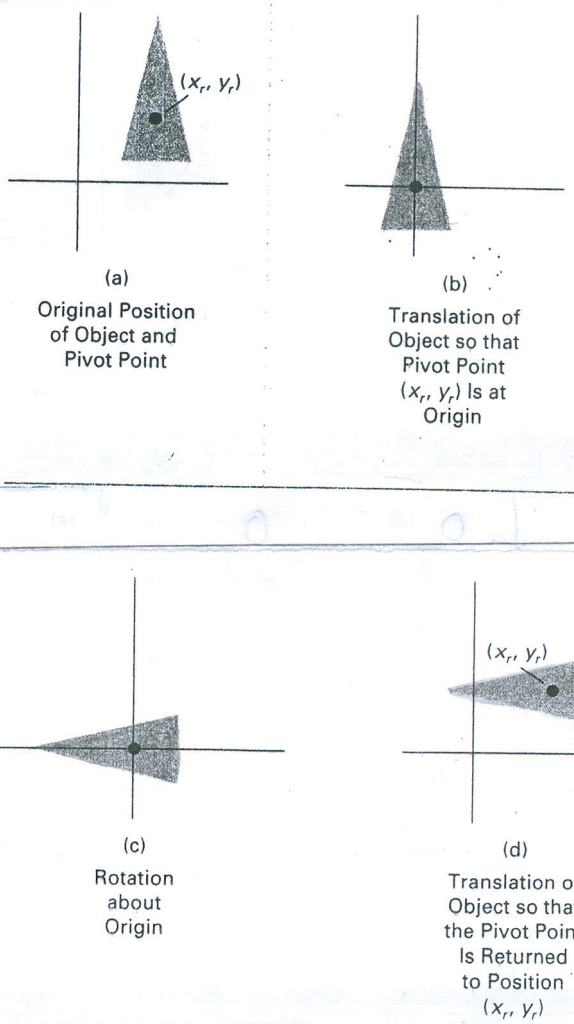


Fig: 1.4

A transformation sequence for rotating an object about a specified pivot point using the rotation matrix $\mathbf{R}(\theta)$ of transformation).

The composite transformation for this sequence is obtained with concatenation

$$\begin{bmatrix} 1 & 0 & x_r \\ 0 & 1 & y_r \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & -x_r \\ 0 & 1 & -y_r \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \cos\theta & -\sin\theta & x_r(1-\cos\theta)+y_r\sin\theta \\ \sin\theta & \cos\theta & y_r(1-\cos\theta)-x_r\sin\theta \\ 0 & 0 & 1 \end{bmatrix}$$

which can be expressed in form:

$$T(x_r, y_r) \cdot R(\theta) \cdot T(-x_r, -y_r) = R(x_r, y_r, \theta)$$

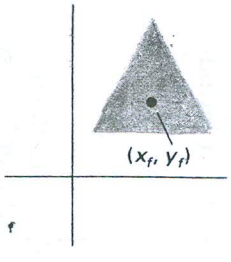
GENERAL FIXED POINT SCALING

To scale an object about a fixed point (x_f, y_f) we perform following sequence of operations:

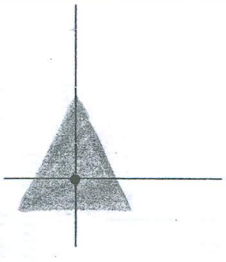
1) Translate object so that the fixed point coincides with the coordinate origin

2) Scale the object with respect to the coordinate origin

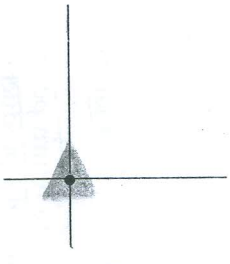
3) Inverse translate the object to return to its original position



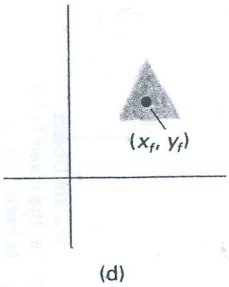
(a)
Original Position of Object and Fixed Point



(b)
Translate Object so that Fixed Point (x_f, y_f) Is at Origin



(c)
Scale Object with Respect to Origin



(d)
Translate Object so that the Fixed Point Is Returned to Position (x_f, y_f)

Fig 5.15
A transformation sequence for scaling an object with respect to a specified fixed position using the scaling matrix $S(s_x, s_y)$ of transformation .

The transformation sequence is illustrated in fig 5.

The composite transformation matrix for this sequence is obtained as:

$$\begin{bmatrix} 1 & 0 & x_f \\ 0 & 1 & y_f \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & -x_f \\ 0 & 1 & -y_f \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} s_x & 0 & x_f(1-s_x) \\ 0 & s_y & y_f(1-s_y) \\ 0 & 0 & 1 \end{bmatrix}$$

which can be expressed in the form

$$T(x_f, y_f) \cdot S(s_x, s_y) \cdot T(-x_f, y_f)$$

$$= S(x_f, y_f, s_x, s_y)$$

This transformation is automatically generated on systems that provide a scale function that accepts coordinates for the fixed point.

REFLECTION OF AN OBJECT ABOUT ANY LINE

When reflection is to be performed about a line that neither passes through origin nor is parallel to coordinate axis can be solved using following five steps:

STEP1: Translate the the line and the object so that the line passes through the origin (T_R)

STEP2: Rotate the line and the object about the origin until the line is coincident with one of the coordinate axis (R_θ)

STEP3: Reflect the object through the coordinate axis (R_{ref})

STEP4: Apply inverse rotation about origin to shift the line at translated position ($R_{(-\theta)}$)

STEP5: Apply inverse translation to

7

send back the object (i.e line) to its original position ($T_{(R)}$) (Fig 16)

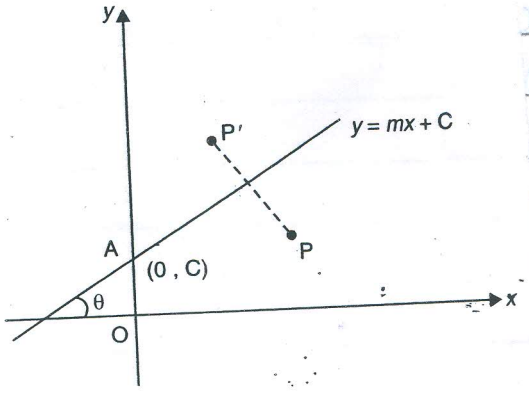


Fig. 16 . Reflection of P about a line.

Combining all these steps we get:

$$[T] = [T_R] \cdot [R_\theta] \cdot [R_{ref}] \cdot [R_\theta]^{-1} [T_R]^{-1}$$

where

T_R = Translation matrix
R_θ = Rotation matrix by θ
R_{ref} = reflection about any axis
$(R_\theta)^{-1}$ = Inverse relation
$[T_R]^{-1}$ = Inverse translation.

Here, $T_R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & c \\ 0 & 0 & 1 \end{bmatrix}$

$$R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_{ref} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_\theta^{-1} = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$T_R^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -c \\ 0 & 0 & 1 \end{bmatrix}$$

Also, slope $= m = \tan\theta$

$$\sin\theta = \frac{m}{\sqrt{m^2+1}}$$

$$\cos\theta = \frac{1}{\sqrt{m^2+1}}$$

Combining all these matrices we get :-

$$[T] = \begin{bmatrix} \frac{1-m^2}{m^2+1} & \frac{2m}{m^2+1} & \frac{-2cm}{m^2+1} \\ \frac{2m}{m^2+1} & \frac{m^2-1}{m^2+1} & \frac{2c}{m^2+1} \\ 0 & 0 & 1 \end{bmatrix}$$

THE WINDOW-TO-VIEWPORT TRANSFORMATION

Much like what we see in real life through a small window on the wall, a computer generated image often depicts a partial view of a large scene.

Objects are placed into a scene by modelling transformation to a master coordinate system, commonly referred to as the world coordinate system (WCS).

A rectangular area with its edges parallel to the axes of WCS is used to select the portion of the scene for which an image is to be generated (fig 17)

This rectangular area is called a window

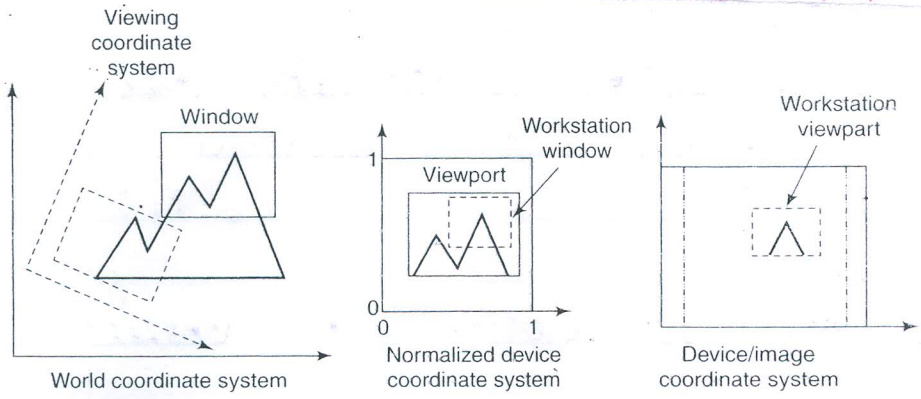


Fig. 11. *Viewing Transformation*

Sometimes an additional coordinate system called viewing coordinate system is introduced to simulate the effect of moving or tilting camera.

On the other hand, an image representing a view of ~~the~~ becomes part of a larger image, like a photo on an album page, which models a computer monitor's display area. We introduce a device independent tool to describe the display area. This tool is called normalized device coordinate system (NDCS).

The process that converts object coordinates in WCS to normalized device coordinates is called Window to viewport mapping or normalized transformation.

WINDOW-TO-VIEWPORT MAPPING

A window is specified by four world coordinates: x_{min}, y_{min}, x_{max} and y_{max}. (Fig 18)

Similarly a viewport is described by four normalized device coordinate u_{min}, v_{min}, u_{max} and v_{max}.

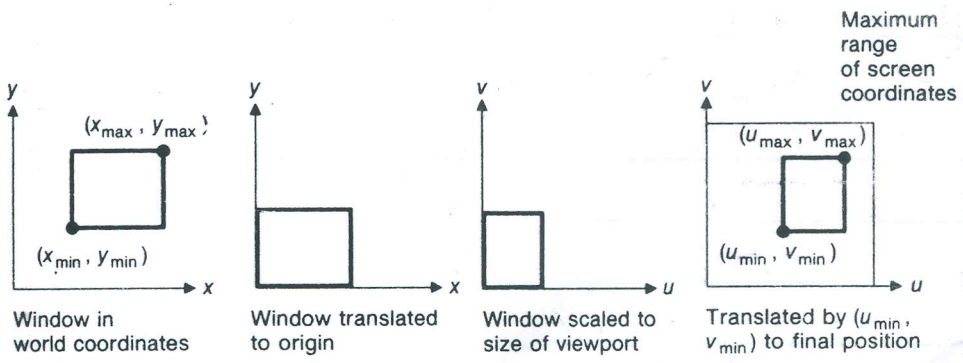


Fig. 18 The steps in transforming a world-coordinate window into a viewport.

The objective of window-to-viewport mapping is to convert the world coordinates (x, y) of an

arbitrary point to its correspondy normalized device coordinates (u, v)

Given a window and viewport, the transformation matrix that maps the window from world coordinates into viewport in device (screen) coordinates can be developed as a three step transformation shown in fig 18

1) The window, specified by its lower left and upper right coordinates, is first translated to the origin of world coordinates

2) Next, the size of window is scaled equal to the size of the viewport

3) Finally, a translation is used to position the viewport.

The overall matrix M_{wv} is shown below.

$$M_{wv} = T(u_{min}, v_{min}) \cdot S\left(\frac{u_{max} - u_{min}}{x_{max} - x_{min}}, \frac{v_{max} - v_{min}}{y_{max} - y_{min}}\right) \cdot T(-x_{min}, -y_{min})$$

$$= \begin{bmatrix} 1 & 0 & u_{min} \\ 0 & 1 & v_{min} \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} \frac{u_{max} - u_{min}}{x_{max} - x_{min}} & 0 & 0 \\ 0 & \frac{v_{max} - v_{min}}{y_{max} - y_{min}} & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & -x_{min} \\ 0 & 1 & -y_{min} \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{u_{max} - u_{min}}{x_{max} - x_{min}} & 0 & -x_{min} \cdot \frac{u_{max} - u_{min}}{x_{max} - x_{min}} + u_{min} \\ 0 & \frac{v_{max} - v_{min}}{y_{max} - y_{min}} & -y_{min} \cdot \frac{v_{max} - v_{min}}{y_{max} - y_{min}} + v_{min} \\ 0 & 0 & 1 \end{bmatrix}.$$

Multiplying $P = M_{wv}\,[x \quad y \quad 1]^T$ gives the expected result:

$$P = \left[(x - x_{min}) \cdot \frac{u_{max} - u_{min}}{x_{max} - x_{min}} + u_{min} \quad (y - y_{min}) \cdot \frac{v_{max} - v_{min}}{y_{max} - y_{min}} + v_{min} \quad 1\right].$$

EFFICIENCY

* The most general composition of R, S & T operation produces a matrix of the form

$$M = \begin{bmatrix} r_{11} & r_{12} & t_x \\ r_{21} & r_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix} \quad - (1)$$

* The upper 2x2 sub matrix is a composite rotation and scale matrix whereas t_x and t_y

are composite translation.

* Calculating M.P @ a vector multiplied by 3×3 matrix takes 9 multiplies and 6 adds. The fixed structure of row of eq ① however simplifies the actual operations:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & t_x \\ r_{21} & r_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \quad -②$$

gives

$$x' = x \cdot r_{11} + y \cdot r_{12} + t_x$$
$$y' = x \cdot r_{21} + y \cdot r_{22} + t_y$$

reducing the process to 4 multiplies and 4 adds - a significant speedup. specially since the operation can be applied to hundres or even thousands of points per picture.

3D TRANSFORMATIONS

Methods for geometric transformations in 3D are extended from 2D methods by including consideration for the z coordinate.

TRANSLATION

In a 3D homogeneous coordinate representation, a point or object is translated as shown in fig 19 and 20

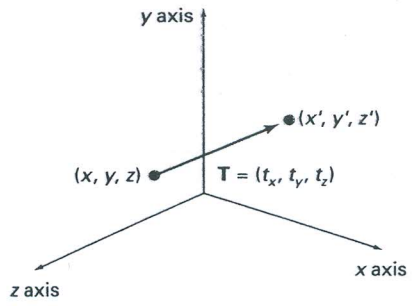


Fig: 19
Translating a point with translation vector $\mathbf{T} = (t_x, t_y, t_z)$.

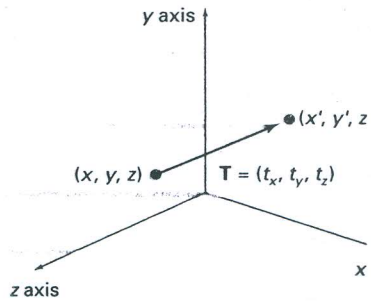


Fig: 20
Translating an object with translation vector **T**.

from position $P = (x, y, z)$ to position $P' = (x', y', z')$ with matrix operation

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

or

$$P' = T \cdot P$$

Parameters t_x, t_y and t_z, specifying the translation distances for the coordinate directions x, y and z.

The above matrix representation is equivalent to three equations

$$x' = x + t_x$$
$$y' = y + t_y$$
$$z' = z + t_z$$

We obtain the inverse of Translation matrix by negating the translation distances t_x, t_y and t_z.

ROTATION

Rotations in 3D is more complex than rotation in 2D.

In 2D, rotation is prescribed by an angle of rotation and a centre of rotation P. In 3D rotation require prescription of an angle of rotation and an axis of rotation (Fig 21)

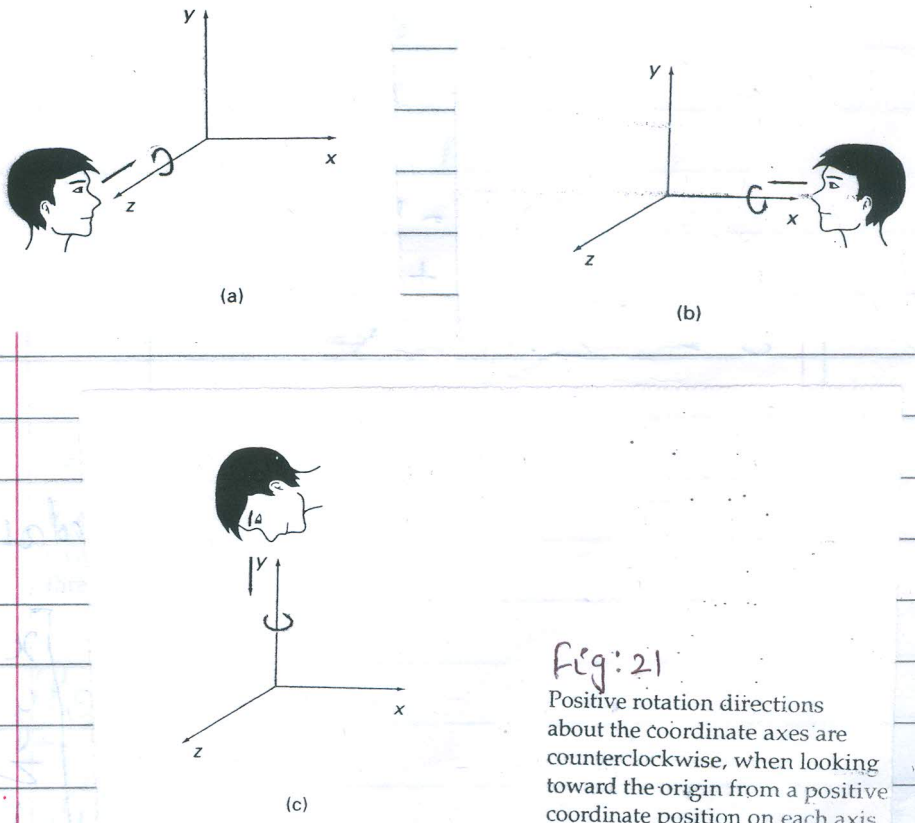


Fig: 21
Positive rotation directions about the coordinate axes are counterclockwise, when looking toward the origin from a positive coordinate position on each axis.

Rotation about z axis

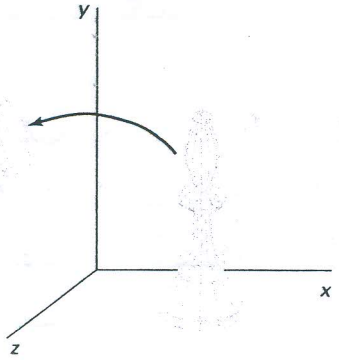


Fig. 22
Rotation of an object about the z axis.

Fig 22 illustrate rotation of an object about the z axis.

The equations are:

$$x' = x\cos\theta - y\sin\theta$$
$$y' = x\sin\theta + y\cos\theta$$
$$z' = z.$$

In homogeneous coordinate system it can be represented as:

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

which can be written as:

$$P' = R_z(\theta) \cdot P$$

The transformation equations for rotation about the other two coordinate axes can be obtained with a cyclic permutation of the coordinate parameters x, y and z in above equations. That is, use the replacements

$$x \rightarrow y \rightarrow z \rightarrow x.$$

as shown in fig 23.

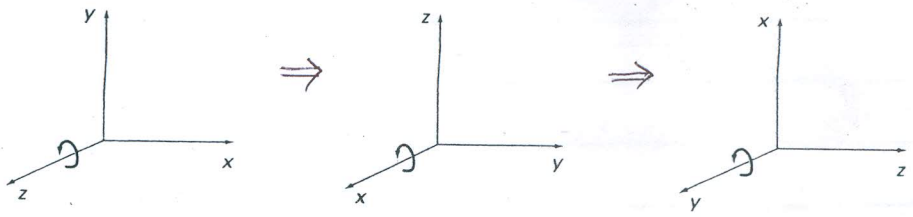


Fig: 23
Cyclic permutation of the Cartesian-coordinate axes to produce the three sets of coordinate-axis rotation equations.

Rotation about x axis

Substituting the permutations we get:

$$y' = y\cos\theta - z\sin\theta$$
$$z' = y\sin\theta + z\cos\theta$$
$$x' = x.$$

which can be written in homogeneous coordinate form as:

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta & 0 \\ 0 & \sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

or

$$P' = R_x(\theta) \cdot P.$$

Rotation about y axis

The transformation equations are:

$$z' = z\cos\theta - x\sin\theta$$
$$x' = z\sin\theta + x\cos\theta$$
$$y' = y$$

The matrix representation for y axis rotation is:

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & 0 & \sin\theta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin\theta & 0 & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

or

$$P' = R_y(\theta) \cdot P$$

An inverse rotation matrix is formed by replacing the rotation angle θ by $(-\theta)$.

Negative values for rotation angles generate rotations in clockwise direction.

SCALING

The matrix expression for the scaling transformation of a position $P = (x, y, z)$ is written as:

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

or

$$P' = S \cdot P$$

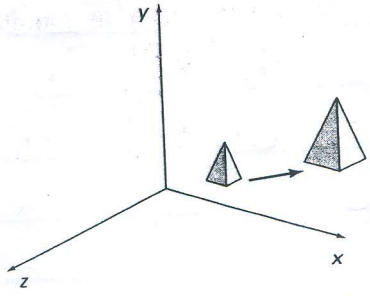


Fig:2.4
Doubling the size of an object with transformation also moves the object farther from the origin.

where scaling parameters s_x, s_y and s_z are assigned positive values.

The equations are:

$$x' = x \cdot s_x$$
$$y' = y \cdot s_y$$
$$z' = z \cdot s_z$$

Fig 2.4 shows scaling an object uniformly.

REFLECTION

A 3D reflection can be performed relative to a selected reflection axis

or with respect to a selected reflection plane.

Case 1: Reflection of points relative to xy plane. The transformation matrix representation is:

$$RF_z = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

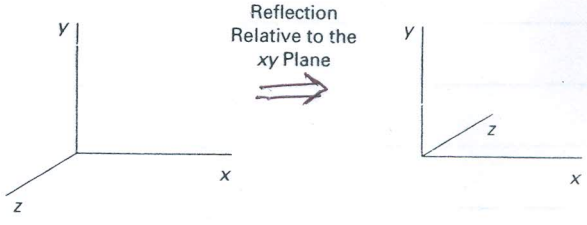


fig: 25
Conversion of coordinate specifications from a right-handed to a left-handed system can be carried out with the reflection transformation 11-46.

SHEARING

Shearing transformation can be used to modify object shape.

As an example of 3D shearing,

the following transformation produces z axis shear.

$$SH_z = \begin{bmatrix} 1 & 0 & a & 0 \\ 0 & 1 & b & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Parameters a and b are assigned any real values.

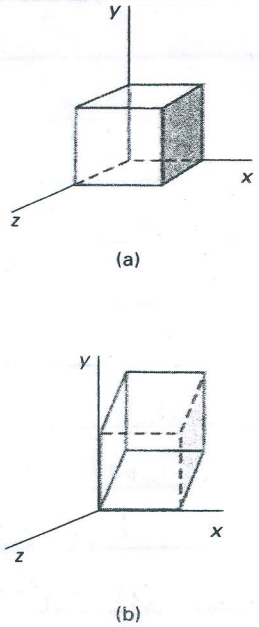


Fig: 2.6
A unit cube (a) is sheared (b) by transformation matrix , with $a = b = 1$.

The above fig 2.6 shows an example of effect of shearing on unit cube for shearing values $a = b = 1$.

TRASFORMATIONS AS A CHANGE IN COORDINATE SYSTEM

Sometimes objects are described in new cartesian frames that take advantage of object symmetries. Coordinate descriptions in these systems must be converted to cartesian device coordinates for display.

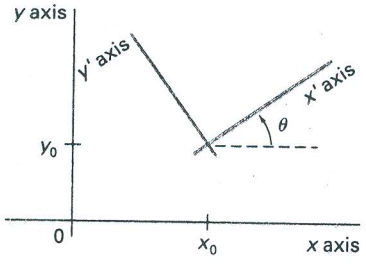


Fig 27
A Cartesian $x'y'$ system positioned at (x_0, y_0) with orientation θ in an xy Cartesian system.

Fig 27 shows two cartesian systems, with their coordinate origin at (0,0) and (x_0, y_0) and with an orientation angle θ between the x and x' axis

To transform object description from xy to x'y' coordinates, we need to set up a transformation that superimposes the x'y' axes

onto the xy axes. This is done in two steps:

1) Translate so that origin (x_0, y_0) of x'y' system is moved to the origin of the xy system.

2) Rotate the x' axis onto the x axis.

Translate the coordinate origin is expressed with the matrix operation:

$$T(-x_0, -y_0) = \begin{bmatrix} 1 & 0 & -x_0 \\ 0 & 1 & -y_0 \\ 0 & 0 & 1 \end{bmatrix}$$

And the orientation of the two systems after translation operation would appear as fig 28

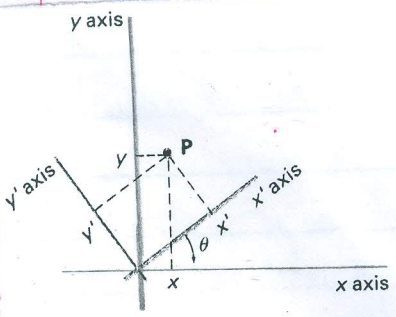


Fig: 28
Position of the reference frames shown in Fig. 5-26 after translating the origin of the $x'y'$ system to the coordinate origin of the xy system.

To get the axes of two systems into coincidence, we then perform clockwise rotation.

$$R(-\theta) = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Concatenating these two transformation matrices gives us complete composite matrix for transforming object description from the xy system to x'y' system.

$$M_{xy,x'y'} = R(-\theta) \cdot T(-x_0, -y_0).$$